Sean D. Lirazan

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EDUCATION

Iowa State University, College of Engineering **GPA: 3.69**

B.S. Mechanical Engineering, Minor in Cyber-Physical Systems *Expected May 2028*

Honors: Tau Beta Pi; Dean's List (F24, S25, F25) | Coursework: Controls, Dynamics, Embedded Systems

WORK EXPERIENCE

NASA Armstrong Flight Research Center | *Controls/Robotics Engineer (Pathways)* Aug 2026–Present | Edwards, CA

- Integrated Simulink-autocoded roll/pitch/yaw rate flight controllers into JPL's F´ (F Prime) C++ flight software framework for the HPSC Cub testbed; verified real-time execution on flight-computer hardware, matching simulation within **2%** across **40+** test cases.
- Debugged and modified C++ sensor-acquisition code for the X-59 supersonic research aircraft, resolving pressure-sensor data-processing and fault detection and handling issues and increasing usable test data by **54%** during outdoor testing.
- Built a rapid-prototyping workflow for X-56 flutter-suppression research, converting proprietary CAD into a scaled 3D-printed model with IMUs and actuated control surfaces; bench tests showed a **35%** reduction in wing-tip oscillation.
- Built MATLAB/Simulink and Python flight-dynamics and control-system simulations for autonomous taxi, takeoff, and landing systems and 737 reduced-gravity flight.

Collins Aerospace (RTX) | *Systems Engineer* Spring 2026–Aug 2026 | Cedar Rapids, IA

- Executed **70+** formal requirements-based verification and validation (V&V) procedures and hardware-in-the-loop (HIL) simulations for CAAS flight software and avionics on 160th SOAR MH-47G and MH-60M aircraft, finding **30+** defects across **3** subsystems.
- Verified display, sensor-fusion, and mission-system software on representative flight hardware, translating software changes into lab tests; worked directly with 160th SOAR stakeholders toward customer acceptance.
- Built **10+** DOORS/DXL scripts automating requirements parsing and test workflows and implemented **15+** process improvements; **4** scripts were adopted team-wide, saving **50+** hours of test-event time.

John Deere | *Robotics & Design Engineer* Summer 2025 | Waterloo, IA

- Prototyped an autonomous precision-agriculture UAV with NDVI and GPS sensing, mapping crop health across **100+** acres.
- Developed GPS and computer-vision sensor-fusion target following with real-time Wi-Fi telemetry, holding tracking error under **1 m**.

ISU Student Innovation Lab | *Embedded Systems & Controls Engineer* Fall 2024–Present | Ames, IA

- Led **5+** embedded C/C++ builds on STM32, Arduino, and Raspberry Pi, including a smart greenhouse, biometric tracker, and Python/Qt arcade machine.
- Mentored **10+** students in sensor integration (IMU, LiDAR, thermal, ultrasonic), microcontrollers, and RTOS.

PROJECTS

Reaction-Wheel Balancing Robot | *ESP32, FreeRTOS, F´, Simulink*

- Derived the nonlinear dynamics, identified motor and inertia parameters from test data, and designed an LQR controller in Simulink; deployed on an ESP32 under FreeRTOS, recovering from **15°** disturbances with under **2°** steady-state error.
- Implemented Kalman-filter state estimation fusing gyro and accelerometer data at **1 kHz**, cutting angle noise **80%** versus raw accelerometer readings; structured the flight software as F´ components with ground-system telemetry.

6-DOF VR-to-Physical Robotic Arm | *3DPD Club | CAD, FEA, ROS 2, NEMA actuators* Ames, IA

- Designed a 6-DOF robotic manipulator with custom 3D-printed planetary NEMA actuators and FEA-validated links.
- Implemented inverse kinematics and real-time VR teleoperation in ROS 2, tracking the operator's hand with under **100 ms** latency and **2 mm** repeatability.
- Added a kinematic-coupling end-effector tool changer with pogo-pin tool ID, completing **100/100** automated tool swaps.

LEADERSHIP & ACTIVITIES

3D Printing & Design Club | *President* Fall 2024–Present | Ames, IA

- Led a **15**-member club through a pivot to robotics and embedded systems, integrating 3D-printed frameworks with microcontroller automation; built partnerships with **10+** companies.
- Delivered an autonomous irrigation system to Reiman Gardens and provided 3D-printing services for local growers.

SKILLS

Robotics & GNC: PID, LQR, state-space control, Extended Kalman filtering, state estimation, sensor fusion, inverse kinematics, ROS 2
Flight & Embedded Software: F´ (F Prime), C/C++, Python, FreeRTOS/RTOS, STM32, ESP32, Arduino, Raspberry Pi, CAN, I2C, Git
Simulation & Test: MATLAB/Simulink, autocode, flight dynamics, hardware-in-the-loop (HIL), V&V, DOORS/DXL, JIRA
Mechanical & Other: SolidWorks (Certified), Onshape, FEA, CFD, 3D printing, rapid prototyping, OpenCV, C#, Java, JavaScript